

Dhruv Maheshwari

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EDUCATION

PES University, Bengaluru

2027

Bachelor of Technology - **Computer Science and Engineering**

Prof. CNR Rao **Scholarship** (40% Tuition Merit Award), **4× Distinction Award**

SKILLS

Languages: Python, JavaScript/TypeScript

Frameworks & Libraries: React, Uvicorn, FastAPI, ChromaDB, FastEmbed, LangChain, Three.js

Tools & Platforms: Git, GitHub, Docker, Kubernetes, AWS Bedrock, Agno, Blender, Vercel, Render, Azure Dev, HuggingFace, Groq

Concepts: API design, Multi-agent AI systems, Retrieval Augmented Generation (RAG), LLM inference, Embeddings, CI/CD

Coursework: DSA, OS, DBMS, Computer Networks, OOP

EXPERIENCE

Intern – Technology

Jun – Aug 2026

Acenet Consulting

Python · AWS Bedrock · FastAPI · Uvicorn · Pandas · NumPy · Azure Deployment · LangChain

- Designed and developed a **multi-agent** AI-powered Virtual CFO platform using Python, FastAPI, and AWS Bedrock to deliver macroeconomic, industry, and automotive market intelligence.
- Built **automated data ingestion, cleaning, and historical analytics pipelines**; integrated LLM-based reasoning agents for executive insights; implemented intelligent query routing across specialized domains.
- Developed **REST APIs**, scenario analysis, and **confidence scoring** with automated reporting, optimizing data processing workflows to support data-driven business decisions.

AI App Development Intern

Jun – Jul 2025

xForm Solutions

Python · PyTorch · NumPy · BERT · ASR · Kaggle

- Worked on a **multilingual speech understanding system** (Whisper + NLLB-200 + SBERT) achieving **sub-0.07 Character Error Rate (CER)** on English benchmark speech with semantic consistency across code-mixed inputs.
- Conducted systematic **WER/CER evaluation** on the Google FLEURS dataset, optimizing the preprocessing pipeline to improve non-English transcription accuracy.
- Engineered **rule-based urgency** and post-processing to improve precision in **noisy speech environments**.

PROJECTS

pRAGma - RAG Application

Python · FastAPI · React (Vite) · Prompt Engineering · FastEmbed · Groq API

github.com/dhruvm-04/pRAGma

- Built a **full-stack RAG application** that lets users upload text PDFs and ask grounded questions, with ingestion for **parsing, chunking, and local embedding generation**.
- Implemented **cosine-similarity retrieval** that exposes **retrieved chunks, similarity scores, source snippets**, latency, and token usage for **transparent, explainable** answers.
- Added multiple **query-execution strategies: Direct, Multi Query, RAG Fusion, Decomposition, Step Back, and HyDE**; with a UI that **visualizes each mode's execution pipeline** and retrieval steps.

aiCFO - Multi Agent Advisory Application

Python · LangChain · LangGraph · Groq · Flask · Pandas · yFinance github.com/dhruvm-04/aiCFO

- Engineered a **parallel multi-agent workflow** using **LangGraph**, routing a single query to specialized Automotive and Macroeconomic agents before synthesis by a CFO agent.
- Implemented **LLM-driven tool calling** across government automotive and economic datasets, enabling agents to **dynamically select domain-specific tools** for trend and metric analysis.
- Built **schema-constrained agent communication** and strict JSON-schema outputs, producing typed analyses with findings, trends, data points, caveats, and recommendations.

MusiConvert - Secure Peer-to-Peer Playlist Sharing Platform

Python · Socket Programming · Fernet Encryption · Multithreading github.com/dhruvm-04/MusiConvert

- Implemented a **custom Selective RepeatARQ protocol** from scratch over **raw UDP sockets** - per-packet **sequencing**, **sliding windows**, threaded **retransmission timers**, and cumulative **ACKs** - to reliably transfer playlists directly between peers on a local network.
- Designed a fuzzy-matching engine that scores candidate tracks using weighted title/artist string similarity plus a duration-tolerance check, so mismatched metadata across platforms **doesn't produce false matches**.

CERTIFICATIONS [Link](#)

- | | |
|--|-----------------|
| • Google IT Automation with Python | <i>Jun 2023</i> |
| • AWS Prompt Engineering Foundation | <i>Mar 2025</i> |
| • IBM Fundamentals of AI | <i>Jan 2025</i> |

EXTRACURRICULARS

- Taught **3D modeling** and **game-ready asset creation** to ~50 students as **Blender Workshop Instructor**, PES University.
- **Led chassis design and fabrication** for the automotive club **Team Vegavath** using Blender and Fusion360.
- Served as **Under Secretary General** and led the Design team for PES University's **Model United Nations society MUNSoc PESUECC**.
- **Led the Design team** for **Maaya 2025**, PES University's **annual techno-cultural fest**.